Josh Visser

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Research interests: Applied electromagnetics; RF and millimeter-wave sensing; radar signal processing; machine learning for physical systems; electromagnetic simulation and digital twins.

Education

University of Waterloo — Master of Applied Science May 2026 – Present
Electrical and Computer Engineering | Expected graduation: May 2028
GPA: 4.0/4.0 (96%) | Wireless Sensors and Devices Lab | Supervisor: Prof. George Shaker
Coursework: Machine Learning; Radio-Wave Systems; Microwave and RF Engineering; Radiation and Propagation of Electromagnetic Fields.

University of Guelph — Bachelor of Engineering Sept 2020 – May 2025
Biomedical Engineering | GPA: 3.71/4.0 (83.9%)
Coursework: Signal Processing; Systems and Control; Probability and Statistics; Biomedical Signals; Bio-Instrumentation Design.

Research Experience

Graduate Researcher — University of Waterloo May 2026 – Present
Wireless Sensors and Devices Lab

- Develop radar simulation and data-analysis tools using Python and Ansys SBR+/Perceive EM, connecting animated human motion and electromagnetic scenes to synthetic radar measurements.
- Build processing pipelines for simulated and measured ADC data, including clutter suppression, range–Doppler processing, and azimuth/elevation feature extraction for gesture recognition.
- Evaluate real-only, synthetic/real, and domain-adaptation learning approaches for simulation-to-real transfer.
- Develop and evaluate fixed-state RIS-assisted 60 GHz sensing for human presence detection, tracking, and touchless gesture recognition beyond a forward-looking radar's field of view.
- Conduct Infineon BGT60TR13C experiments comparing RIS, metal-reflector, and radar-only configurations; analyze range profiles, detection contrast, and angle estimates.

Machine Learning and Radar Research — University of Waterloo May 2024 – Jan 2025
Wireless Sensors and Devices Lab

- Developed a privacy-preserving 60 GHz FMCW radar system for activity monitoring in long-term-care bathrooms, combining radar signal processing and deep learning.
- Built preprocessing and range–Doppler/range–elevation feature pipelines; achieved approximately 95% activity-classification accuracy in evaluated conditions.
- Led experimental data collection with 10+ participants and evaluated sensing performance across bathroom configurations.

Selected Technical Projects

Multi-Radar Tracking System with Modified GRIN Lens 2025

- Developed Python multiprocessing acquisition, baseline calibration, zone-based detection logic, and a live GUI for five FMCW radar modules in a lens-assisted system with 140-degree angular coverage.
- Integrated radar hardware and software with the team and contributed to experimental validation. Presented the system at the IEEE AP-S Student Design Contest; team placed third.

AI Medication Assistant and Smart Pillbox — University of Guelph Sept 2024 – Apr 2025

- Prototyped a computer-vision pillbox for medication-adherence verification and an LLM-based interface for medication information, symptom reporting, and dosage-compliance support.

Teaching Experience

Teaching Assistant — University of Waterloo Fall 2026
Probability and Statistics (ECE 203).

Publications

Journal article

1. **J. Visser**, H. Abedi, and G. Shaker, "Activity Monitoring System for Long-Term Care Bathrooms Using mm-Wave FMCW Radar," *IEEE Sensors Journal*, vol. 26, no. 14, pp. 21719–21735, 2026. [DOI].

Conference papers

2. M. O. Bagheri, V. Leong, J. Chow, **J. Visser**, and G. Shaker, "Modified GRIN Luneburg Lens for High-Efficiency Millimeter-Wave Multi-Radar Sensing," *European Conference on Antennas and Propagation (EuCAP)*, pp. 1–3, 2026. [DOI].
3. **J. Visser**, H. Abedi, S. Karmani, and G. Shaker, "AI-Based Activity Monitoring in Washrooms Using Low-Resolution Radar," *IEEE Latin American Microwave Conference (LAMC)*, 2025. [DOI].
4. J. Hall, J. B. Bae, **J. Visser**, H. Abedi, and G. Shaker, "Validating mm-Wave Radar System for Fall Detection in Washrooms," *IEEE Healthcare Innovations and Point-of-Care Technologies (HI-POCT)*, 2024. [DOI].

Preprint and submitted work

- M. O. Bagheri, J. Chow, **J. Visser**, V. Leong, and G. Shaker, "Millimeter-Wave Multi-Radar Tracking System Enabled by a Modified GRIN Luneburg Lens for Real-Time Healthcare Monitoring," 2026. arXiv:2601.12629. Submitted to *IEEE Transactions on Antennas and Propagation*.

Presentations

IEEE AP-S Student Design Contest — Ottawa, Canada July 2025
Presented the lens-assisted multi-radar tracking system to judges and peers.

IEEE LAMC — San Juan, Puerto Rico Jan 2025
Presented radar activity-monitoring research to 30+ attendees.

Industry Experience

Hardware Engineer — Evertz Microsystems, Burlington, ON Jan – May 2025; May – Aug 2022

- Designed and tested high-speed digital PCBs; validated power-integrity simulations with hardware measurements and resolved 30+ component-sourcing and power-supply design issues.

R&D Engineer — Accelovant Technologies, North Vancouver, BC Jan – Aug 2023

- Developed custom SFP modules and temperature probes for fiber-optic sensing; designed a bed-of-nails test fixture supporting quality control for 5,000+ PCBs annually.
- Automated generation of 50+ SolidWorks drawings using a Visual Basic macro.

Awards and Scholarships

NSERC Canada Graduate Research Scholarship—Master's (CGRS-M), 2026
President's Graduate Scholarship, University of Waterloo, 2026
Third Place, IEEE AP-S Student Design Contest (team award), 2025
Rising Stars Research Fellow, University of Waterloo, 2024
Dean's Honour List, University of Guelph, all semesters, 2020–2025

Technical Skills

Programming and ML: Python (NumPy, PyTorch, TensorFlow/Keras, OpenCV, scikit-learn), MATLAB, C (basic), Visual Basic.
Radar and simulation: FMCW radar; FFT processing; MTI; angle estimation; domain adaptation; Ansys SBR+/Perceive EM.
Hardware and design: PCB design and testing, power integrity, circuit debugging, Altium, LTspice, SolidWorks, soldering, 3D printing.

Leadership and Volunteering

University of Guelph Robotics Team — Mechanical Engineer, Sept 2022–May 2024. Led robotic-arm redesign for the Canadian International Rover Competition.
University of Guelph — Engineering Peer Helper, Sept 2021–May 2022. Mentored students in organization, teamwork, and time management.